

Today we explore $I^G (-5P_C)$

conservation laws

NB. I always conserved

	P	C	I	G
STRONG	✓	✓	✓	✓
ELM	✓	✓	X	X
WEAK	X	X	X	X

But let's see all these in detail!

① $m \rightarrow pe$ why forbidden even w/o postulating LNC

$m = \text{udd} \rightarrow \text{neutral baryon}$

What is $J(m) = ?$

• 3 quarks in s-wave $\rightarrow L = 0$

• $S_{123} \begin{matrix} \uparrow\uparrow\uparrow \\ \uparrow\downarrow\uparrow \end{matrix} \Rightarrow \frac{1}{2} \otimes \frac{1}{2} = 0 \oplus 1$
 $\downarrow$
 $0 \otimes \frac{1}{2} = \frac{1}{2} ; 1 \otimes \frac{1}{2} = \frac{1}{2} \oplus \frac{3}{2}$

$$= S_{23} \begin{pmatrix} \frac{1}{2} \\ 3 \\ 2 \end{pmatrix} \quad \boxed{J(m) = \frac{1}{2}} \quad \begin{matrix} \swarrow \text{Initial} \\ \searrow \text{same for } p \end{matrix}$$

$$J(p e) \Rightarrow J(p) \otimes J(e) = \frac{1}{2} \otimes \frac{1}{2} \\ = 0 \oplus 1$$

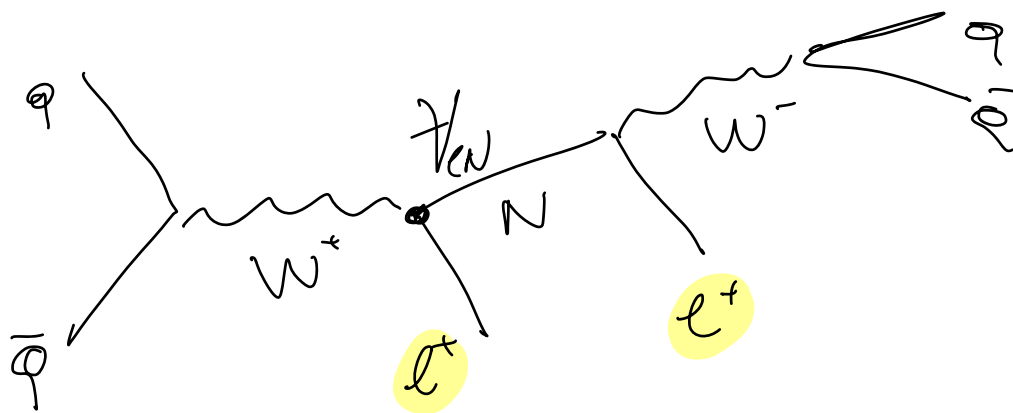
↗
J is not conserved!

But... do we know $\Delta N V$ processes?
Not in SM! But good example
for NP searches:

e.g. heavy Neutral Leptons

$$\begin{array}{c} \text{initial} \\ \hline q\bar{q} \end{array} \rightarrow W^+ \rightarrow \begin{matrix} \swarrow \\ l^+ \end{matrix} \begin{matrix} \nwarrow \\ N \end{matrix} \rightarrow l^+ W^- l^+ \rightarrow \begin{matrix} \text{final} \\ l^+ l^+ \bar{q} q' \end{matrix}$$

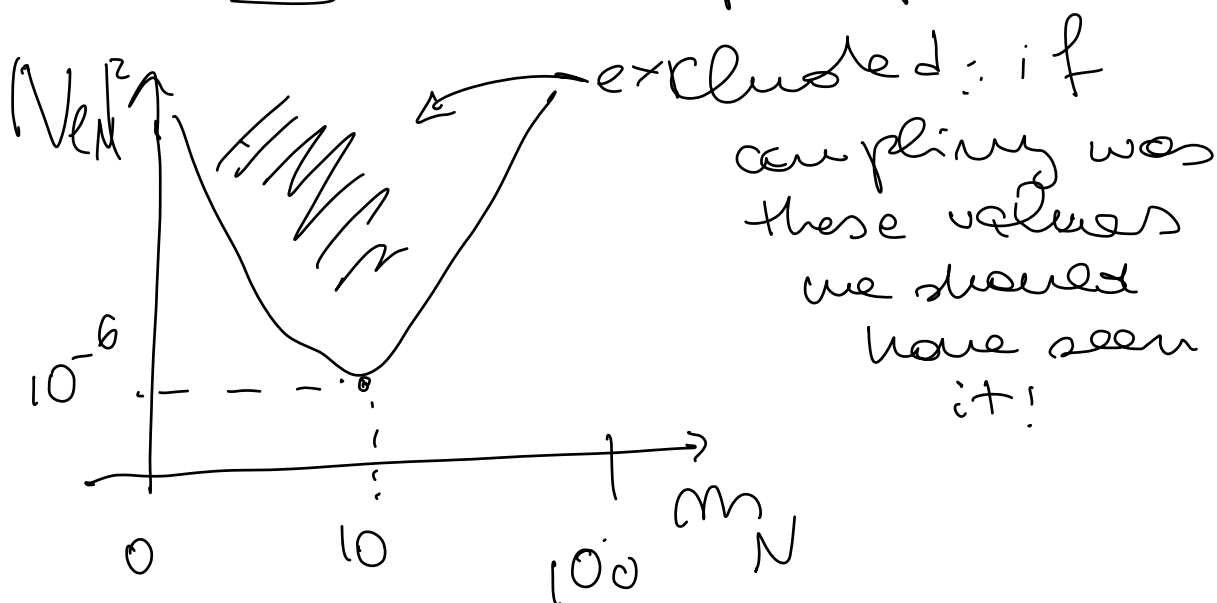
$L=0$ $L=2$



What signature these events leave in the LHC detectors?

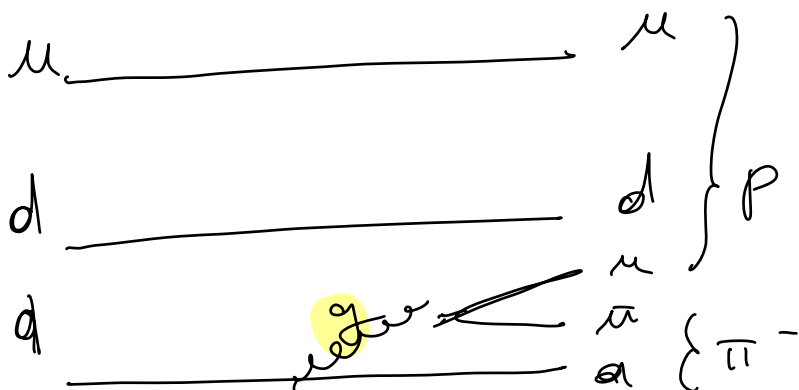
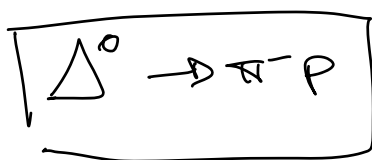
- a jet from $q\bar{q}$
- two leptons $\rightarrow$ SS
- NB N can be prompt or LL

direct search for weird signature



—

② Consider Δ^0 and Λ^0 . Both decay to $\pi^- p$ but very different lifetimes: $\tau_{\Delta^0} \sim 10^{-23} s$ $\tau_{\Lambda^0} \sim 10^{-10} s$



STRONG
DECAY

$$\alpha_s \approx 0.1 - 1$$

$$m_{\Delta^0} = 1232 \text{ MeV}$$

$$m_p = 938 \text{ MeV}$$

$$m_{\pi^-} = 140 \text{ MeV}$$

NB. $\tau \propto \frac{1}{\Gamma} \Rightarrow \tau = \frac{1}{\Gamma}$

$$\Gamma \propto |M|^2$$

↖ interaction strength

final states

light decay products
↓
large Γ

$p \nearrow \quad \pi \nearrow \quad \pi \searrow$

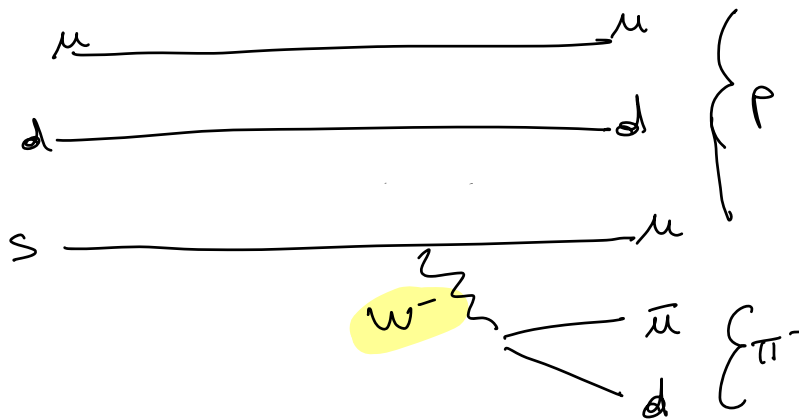
$$\Gamma(\Delta^0) \sim 100 \text{ MeV}$$

$$\tau = \frac{\hbar}{\Gamma} = \frac{6.6 \cdot 10^{-22} \text{ MeV s}}{100 \text{ MeV}} \sim 10^{-23} \text{ s}$$

How do I see experimentally it?

Broad resonance in $\pi\pi$ invariant mass spectrum.

$$\Lambda^0 \rightarrow \pi^- p$$



$$m_{\Lambda^0} \sim 1115 \text{ MeV}$$

WEAK
DECAY

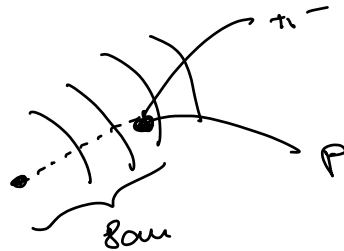
$\downarrow$
b/c s not
conserved

NB

u c t
d $\nearrow$ s b

Smaller transition prob. implies smaller Γ
hence longer τ .

Experimentally $cr(\lambda) \sim 8\mu m$ & i see it as a displaced vertex in the inner tracker detector (V-particle)



③ Analyze decays of $\gamma, \rho, \omega, \phi$ mesons in 2π or 3π based on conservation laws.

$$SU(2) \rightarrow \begin{pmatrix} u \\ d \end{pmatrix} \quad I = \frac{1}{2} \quad I_3 \begin{matrix} +\frac{1}{2} \\ -\frac{1}{2} \end{matrix}$$

s, c, b have $I = 0$

$$SU(3)_F \rightarrow \begin{pmatrix} u \\ d \\ s \end{pmatrix} \quad \begin{matrix} \text{fundam. representation} \\ \uparrow \\ \text{approx. symmetry b/c} \\ m_s \neq m_{ud} \end{matrix}$$

Particles $\begin{cases} \text{mesons } (q\bar{q}) \\ \text{baryons } (qqq) \end{cases}$ can be built by combining u, d, s in the appropriate way.

NB I conserved in STRONG. Now we
can explain why $\Lambda^0 \rightarrow \pi^- p$ is WEAK

$$I(\wedge^0) = \mathcal{O}$$

$$I(\pi) = 1$$

$$I(\rho) = \frac{1}{2}$$

$$I(\pi) = 1 \quad \frac{1}{2} \otimes 1 = \frac{1}{2} \otimes \frac{3}{2} \neq 0$$

I not conserved!

But let's demonstrate these I!

let's build $\downarrow$ the I multiplets.

MESONS $\leftarrow$ b/c $\gamma, \rho, \omega, \phi$ are mesons

$\begin{pmatrix} u \\ d \\ s \end{pmatrix} \otimes \begin{pmatrix} \bar{u} \\ \bar{d} \\ \bar{s} \end{pmatrix} \rightarrow \text{find } I^G(j^{PC})$
 $\swarrow$ pseudo scalar

• J? $L=0, 1, 2 \dots$
 $S=0, 1$

$\rightarrow J \begin{matrix} \swarrow 0 \\ \searrow 1 \end{matrix} \leftarrow \text{vector}$

P? $P(q\bar{q}) = \underbrace{P(q)P(\bar{q})}_{-1} (-1)^L = (-1)^{L+1}$
 $\nwarrow$ orbital

if $L=0 \Rightarrow P=-1$

hence multiplet can be 3^P

• $C?$ ← NB defined only for neutral states

$$C(e\bar{e}) = (-1)^{L+S}$$

if $L=0 \Rightarrow C = (-1)^S$

$$\begin{cases} +1 & \text{if } S=0 \\ -1 & \text{if } S=1 \end{cases}$$

hence multiplets can be

$$3^{PC} \begin{cases} 0^{-+} \\ 1^{--} \end{cases}$$

NB Why in C appears S ?

→ $C \equiv P \otimes \text{Spin flip}$

eg. $e^+e^- : \begin{matrix} \leftarrow \end{matrix} \xrightarrow{\quad} \begin{matrix} \rightarrow \end{matrix} \xrightarrow{C} \begin{matrix} \leftarrow \end{matrix} \xrightarrow{\quad} \begin{matrix} \rightarrow \end{matrix}$

↓ P

$e^- \xleftarrow{\quad} \xrightarrow{\quad} e^+ \xrightarrow{\text{Spin flip}} e^- \xleftarrow{\quad} \xrightarrow{\quad} e^+$

• $I?$ $\begin{pmatrix} u \\ d \end{pmatrix} \rightarrow \frac{1}{2} \otimes \frac{1}{2} = 0 \oplus 1$

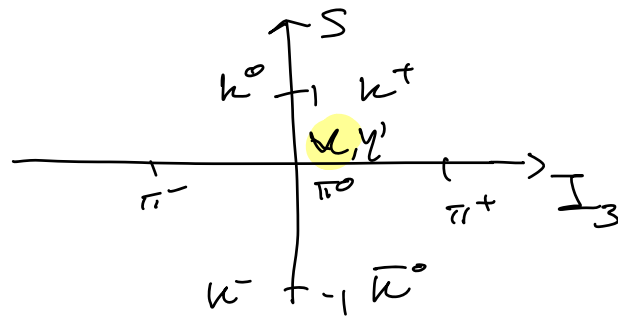
$S \rightarrow 0$ when I combine $u/\bar{u}$ or $d/\bar{d}$

$$\frac{1}{2} \otimes 0 = \frac{1}{2}$$

I can be: $0, \frac{1}{2}, 1$

singlet ← 0 doublet ← $\frac{1}{2}$ triplet ← 1

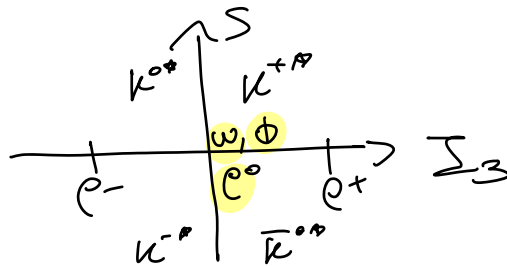
$$J^{PC} = 0^{-+}$$



π_S are a triplet

K_S are doublets

$$J^{PC} = 1^{--}$$



ρ_S are triplet

G ? $\leftarrow$ defined only for multiplets w/ $\sum_i p_i = 0$
(π_S yes, K_S no)

$$G = C e^{i\pi I_2} = (-1)^{L+S} (-1)^I$$

$\leftarrow$ of the neutral element of the multiplets

	$q\bar{q}$	π_S	ρ_S
P	$(-1)^{L+1}$	-1	-1
C	$(-1)^{L+S}$	$+1$	-1
G	$(-1)^{L+S+I}$	-1	$+1$

What happens for a system of n pions?

$$P(n\pi) = (-1)^n (-1)^L$$

$$C(m\pi) = (+1)^m (-1)^L$$

$$G(m\pi) = (-1)^m$$

Now

	$I^G(J^{PC})$	Contains S ?	Mass [MeV]	Γ [MeV]
η	$0^+(0^{-+})$	X	547	0.0013
ρ	$1^+(1^{--})$	X	770	150
ω	$0^-(1^{--})$	X	782	8
ϕ	$0^-(1^{--})$	✓	1020	4

$$\begin{aligned} \eta &\rightarrow \pi^0 \pi^0 < 10^{-9} \\ \eta &\rightarrow \pi^+ \pi^- < 10^{-6} \end{aligned}$$

NB. $\pi \rightarrow 1^-(0^{-+})$

η has $S=0 \rightarrow$ 2 pions in $L=0$

$$P(2\pi) = (-1)^2 (-1)^0 = +1$$

$$C(2\pi) = (-1)^0 = +1$$

$$G(2\pi) = (-1)^2 = +1$$

But η is $0^+(0^{-+})$ hence P is not

conserved and STRONG/EM suppressed!
 in principle can be WEAK but BR $\sim 10^{-17}$

$$\boxed{\begin{array}{l} \gamma \rightarrow 3\pi^0 \\ \gamma \rightarrow \pi^+\pi^-\pi^0 \end{array}}$$

$\leftarrow 3\pi$ in S-wave

$$P(3\pi) = -1$$

$$C(3\pi) = +1$$

$$G(3\pi) = -1$$

In this case G is not conserved, hence
 STRONG is suppressed but EM ok.

$$\text{NB } \Gamma_{\text{EM}} > \Gamma_{\text{STRONG}} \rightarrow \Gamma_{\text{EM}} < \Gamma_{\text{STR.}}$$

this $\nearrow$ explain the narrow width.

$$\boxed{\rho \rightarrow \pi^+\pi^- \sim 100\%}$$

To make it happen since $\rho \rightarrow 1^+ \left(\underset{\uparrow}{1}^{--} \right)$, the 2π s
 have to be in $J=1 \rightarrow L=1$

$$\text{hence } P(2\pi) = (-1)^2 (-1)^L = -1$$

$$C(2\pi) = (-1)^L = -1$$

$$G(2\pi) = (-1)^m = +1$$

can be STRONG

But what about $\rho \rightarrow \pi^0 \pi^0$?

With $S=1$ the wavefunction of the 2π is antisymm. But this violates the B-E principle for a system of identical particles.

Same story for $\rho \rightarrow \pi^0 \pi^0 \pi^0$

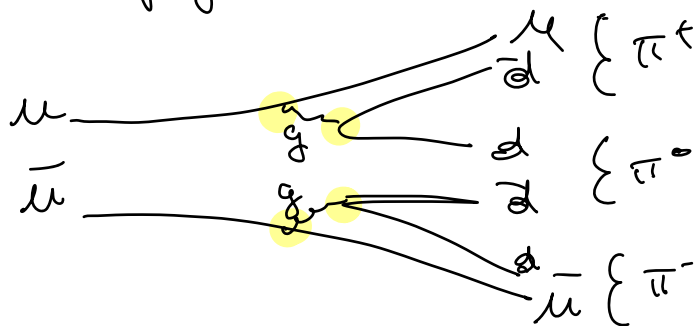
$$\rho \rightarrow \pi^+ \pi^- \pi^0 \sim 10^{-4} \quad \text{vs} \quad \omega \rightarrow \pi^+ \pi^- \pi^0 \sim 90\% \quad \text{vs} \quad \phi \rightarrow \pi^+ \pi^- \pi^0 \sim 15\%$$

why?

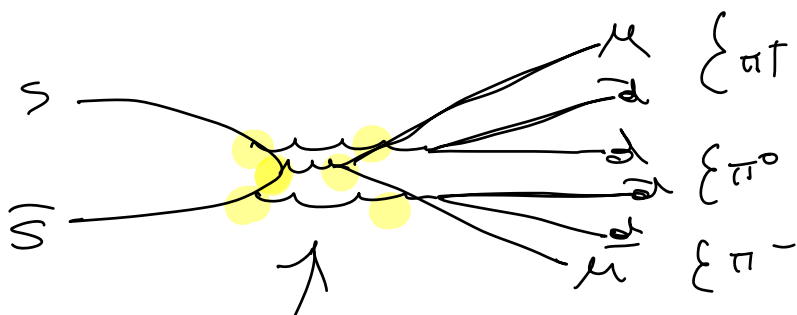
$$G: +1 \neq -1 \quad -1 = -1 \quad -1 = -1$$

But why the difference between ω and ϕ ?

Recall that ϕ has s quarks! Let's look at the Feynman diagrams:



$$\omega \rightarrow \pi^+ \pi^- \pi^0 \sim 90\%$$



$$\phi \rightarrow \pi^+ \pi^- \pi^0 \sim 15\%$$

OZI suppressed process

NB Why 3 gluons? Recall that gluons have color charge.

$$C(g) = -1 \quad C(gg) = +1$$

$1g \rightarrow$ VIOLATES C

$2g \rightarrow$ VIOLATES C

$3g \rightarrow$ ok $\checkmark$

NB Why $C(gg) = +1$

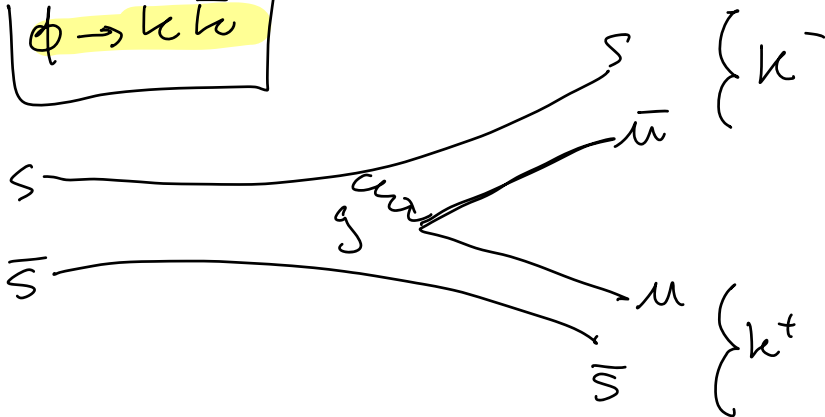
$C = (-1)^{L+S}$ of a system of two identical bosons $\begin{smallmatrix} 0 \\ 2 \end{smallmatrix}$

NB OZI happens also for J/ψ . But why $\Gamma(J/\psi) \sim 92 \text{ keV}$ while $\Gamma(\phi) \sim 4.8 \text{ MeV}$?

$\rightarrow$ This is b/c $\phi \rightarrow K\bar{K}$ can happen ($m_K \sim 495 \text{ MeV}$) while $J/\psi \rightarrow D\bar{D}$ no ($m_D = 1.87 \text{ GeV}$)

④ Why $\phi \rightarrow K\bar{K}$ is the dominant decay?

$$\boxed{\phi \rightarrow K \bar{K}}$$



open strange decay.

⑤ Why $\phi \rightarrow \gamma \gamma$ and $\phi \rightarrow \pi^0 \gamma$ have very different BF?

$$\phi \rightarrow \gamma \gamma \sim 1.3\%$$

$$\phi \rightarrow \pi^0 \gamma \sim 10^{-3}$$

$$m_\phi = 1020 \text{ MeV}$$

$$m_\gamma = 547 \text{ MeV}$$

$$m_{\pi^0} = 135 \text{ MeV}$$

this one should be framed by ΔM .

$$\phi \rightarrow 0^-(1^{--})$$

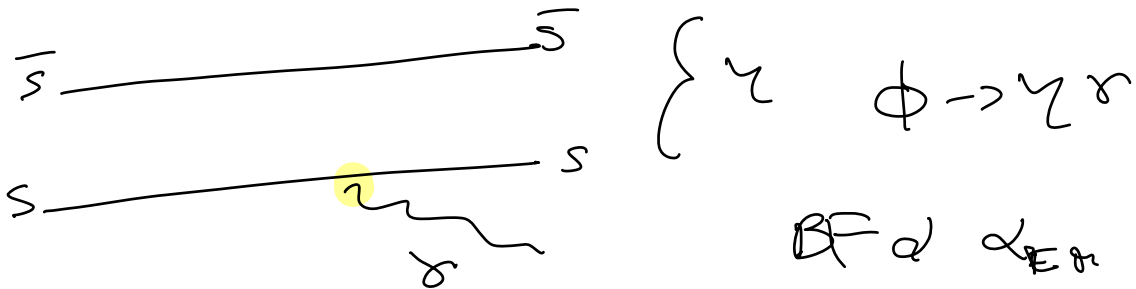
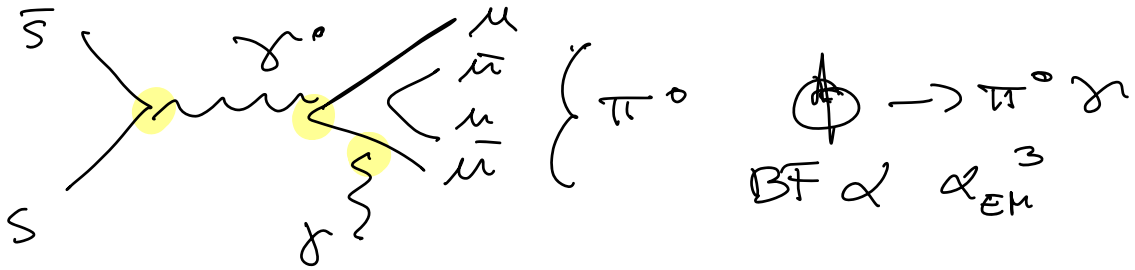
$$\gamma \rightarrow 0^+(0^{-+})$$

$$\pi \rightarrow 1^-(0^{-+})$$

Both are ECM b/c γ does not have I defined, hence I (or G) is not a good quantum number.

Let's look at the diagrams:

Recall that η contains S ...



$\phi \rightarrow \pi^0 \gamma$ more suppressed (the decay that maintains S is more natural)